Effects of Superspreaders in Spread of Epidemic

Spatial SIR model, Monte Carlo simulation, and SARS comparison

Reference: R. Fujie and T. Odagaki, Physica A, 2007

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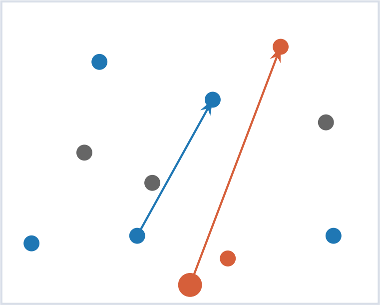


local and long-range transmission paths

Roadmap

1. Problem and modeling approach
2. Spatial SIR model with two superspreader mechanisms
3. Monte Carlo solving algorithm
4. Reproduced simulation results
5. Sensitivity, robustness, and reference conclusion

Problem

- Classical well-mixed SIR models assume similar transmission rates for all infected individuals.
- Real outbreaks often show strong heterogeneity: a small number of cases cause many secondary infections.
- The paper models superspreaders directly and asks which mechanism explains spread better.

Key question

Are superspreaders important because they are more infectious locally, or because they have many social connections?

Modeling Approach

- Individuals are fixed in a two-dimensional continuous domain of side length $L = 10r_0$.
- Each individual has state S, I, or R.
- Transmission probability depends on periodic distance r between an infected individual and a susceptible individual.
- A fraction λ of the population is assigned as superspreaders.

$$\rho = \frac{N}{L^2}, \qquad X_i(t) \in \{S, I, R\}.$$

Two Superspreader Models

Strong infectiousness model

Superspreaders have higher local infection probability but the same cutoff radius:

$$w_s(r) = \begin{cases} w_0, & 0 \le r \le r_0, \\ 0, & r > r_0. \end{cases}$$

Hub model

Superspreaders have a wider contact radius:

$$w_s(r) = \begin{cases} w_0(1 - r/r_n)^2, & 0 \le r \le r_n, \\ 0, & r > r_n, \end{cases}$$

with $r_n = \sqrt{6}r_0$ for superspreaders.

Both mechanisms are normalized to have comparable expected infection strength; their spatial patterns differ.

Basic Reproductive Number and Critical Density

$$R_0(\lambda) = \frac{\rho}{w_0}\left[\lambda \int w_s(r) 2\pi r \, dr + (1-\lambda) \int w_n(r) 2\pi r \, dr\right].$$

$$\rho_c \pi r_0^2 = \frac{R_c}{\lambda + \frac{1-\lambda}{6}}.$$

Strong infectiousness

$R_c \approx 4.5$

Hub model

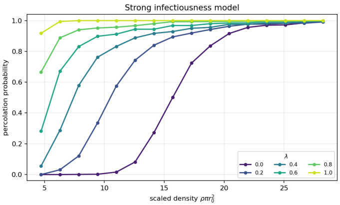
$R_c \approx 3.2$

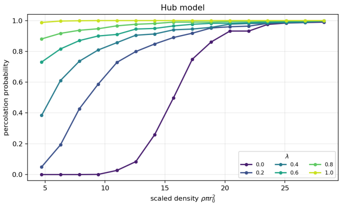
Monte Carlo Solving Algorithm

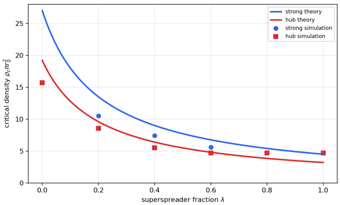
1. Randomly place N individuals in the domain; set one initial infected individual.
2. Randomly assign superspreader labels using fraction λ.
3. At each step, infected individuals attempt to infect susceptible individuals with probability $w(r)$.
4. Newly infected individuals transmit only from the next step.
5. Current infected individuals recover with probability γ.
6. Stop when no infected individuals remain or the maximum number of steps is reached.

Outputs: percolation probability, front distance, propagation velocity, epidemic curve, and secondary infection distribution.

Result 1: Percolation







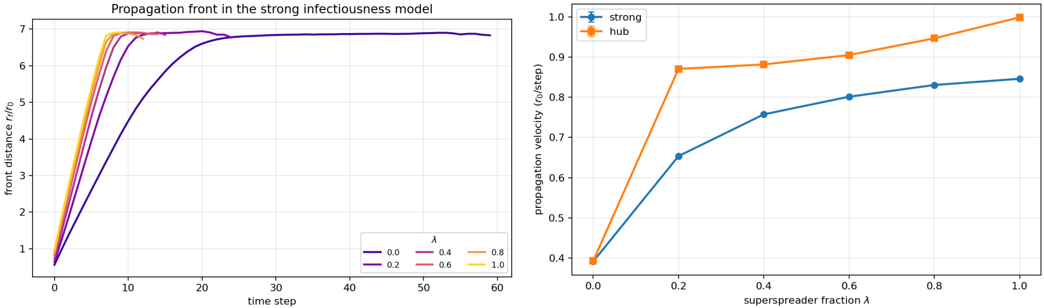
Strong

Hub

Critical density

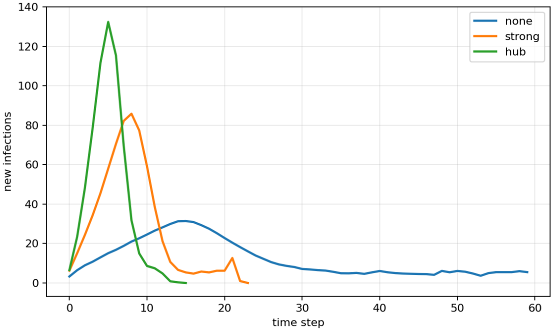
Increasing λ lowers the density threshold for percolation. The hub model percolates at lower density.

Result 2: Propagation Speed



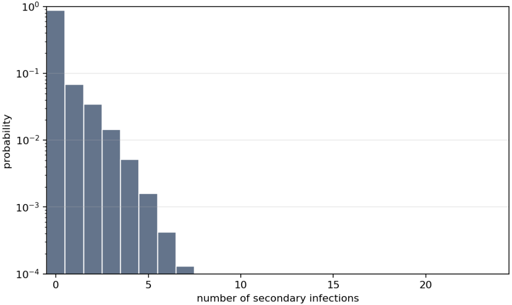
Propagation speed increases with λ. For $\lambda > 0$, the hub model spreads faster because it creates long-range infection paths.

Result 3: Epidemic Curves

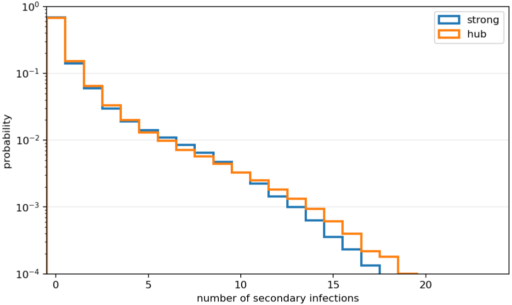


- No-superspreader outbreaks are broader and peak later.
- Superspreaders make the peak earlier and sharper.
- The hub model produces the fastest rise.

Result 4: Secondary Infections



No superspreaders



With superspreaders, $\lambda = 0.2$

Superspreaders create a long-tailed secondary-infection distribution, matching the qualitative pattern discussed in the paper.

Comparison with SARS Data

- The reference paper compares simulations with SARS in Singapore in 2003.
- The observed direct secondary cases include several unusually large values, including 12, 21, 23, and 40.
- The hub model with $N = 477$, $\rho\pi r_0^2 = 15.0$, $\lambda = 0.4$, and $\gamma = 1$ gives the best qualitative epidemic-curve match.

Reference conclusion

SARS superspreading was better explained by many social connections than by stronger local infectiousness alone.

Sensitivity and Robustness

- The model is sensitive to superspreader fraction λ: increasing λ lowers critical density and increases propagation speed.
- It is sensitive to the mechanism of superspreading: hub contacts produce faster spatial spread than stronger local infectiousness.
- Population density controls whether infection chains die out or percolate.
- Recovery probability γ is also important: lower γ means a longer infectious period and more transmission opportunities.

Robust qualitative findings

Across parameter settings, superspreaders lower the percolation threshold, speed up spread, sharpen epidemic curves, and create long-tailed secondary-infection distributions.

Limitations

- Individuals are fixed in space, while real people move and change contacts.
- Superspreader labels are assigned randomly, but real superspreading depends on behavior, occupation, environment, and biology.
- The model is best interpreted as a controlled comparison of two mechanisms, not a full prediction tool for every epidemic.

Conclusion

1. Superspreaders substantially change epidemic dynamics.
2. The hub model spreads faster and has lower critical density than the strong infectiousness model.
3. Long-range social connectivity is the key mechanism in the reference paper's SARS comparison.

Takeaway

Average transmission rate alone is not enough; epidemic models should include heterogeneity in individual connectivity.